

This document provides additional assistance with wiring your Extron IPL PRO enabled product to your device. Different components may require a different wiring scheme than those listed below.

For complete operating instruction, refer to the user's manual for the specific Extron IPL PRO enabled product or the controlled device manufacturer supplied documentation.

Device Specifications:

Device Type: Display
Manufacturer: BenQ
Firmware Version: N/A
Model(s): PL550

Tested on the Following Software and Firmware Versions:

IP Link Pro Control Processor Firmware	Touch Link Panel Firmware	GS Version
2.05.0000-b007	2.01.0001-b001	1.2.3

Version History

Module Version	Date	Notes
1_0_0_0	4/27/2017	Initial Version

Module Notes

- Unidirectional variable must be set to 'True' if status is not required. Default value is 'False'. Example: InterfaceName.Unidirectional = 'True'
- connectionCounter variable must be set the number of queries that will be sent to the device before displaying 'Disconnected' if no response is received. Default value is 15. Example: InterfaceName.connectionCounter = 5
- DisplayID variable must be set accordingly. Default value is '1'. Display ID ranges from '1' to '99' and 'Broadcast'. Example: InterfaceName.DisplayID = '1'
- When DisplayID is set to 'Broadcast', status is unavailable as the module becomes unidirectional
- DisplayID needs to match the ID on the Display. Default ID is set to 1.

Control Commands, States, and Qualifiers:

AspectRatio	'Full' ¹ 'Zoom'	'4:3' ²	'Wide Zoom' ³
AudioMute	'On'	'Off'	
AutoImage ⁴	None		
ExecutiveMode	'On'	'Off'	
['Type']	'Button'	'IR'	'Button and IR'
Input	'VGA' 'YPbPr' 'DisplayPort' 'USB Display'	'HDMI' 'S-Video' 'Multi-Media'	'AV' 'DVI' 'Network'
OnScreenDisplay	'On'	'Off'	
PAPEnable	'Off'	'PIP'	'PBP'
PictureMode	'Standard' 'Custom'	'Vivid'	'Cinema'
PIPPosition	'Upper Left' 'Lower Right'	'Upper Right'	'Lower Left'
Power	'On'	'Off'	
Volume	0 to 100 in steps of 1		

¹ Aspect ratio is Full for video and Full 2 for PC.² Aspect ratio is 4:3 for video and Real for PC.³ Aspect ratio is Wide Zoom for video and Full 1 for PC.⁴ Only applicable for VGA input.

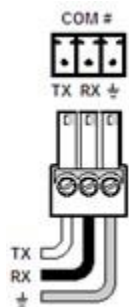
Status Available:

AspectRatio	'Full' ¹ 'Zoom'	'4:3' ²	'Wide Zoom' ³
AudioMute	'On'	'Off'	
ConnectionStatus	'Connected'	'Disconnected'	
ExecutiveMode	'On'	'Off'	
['Type']	'Button'	'IR'	'Button and IR'
Input	'VGA' 'YPbPr' 'DisplayPort' 'USB Display'	'HDMI' 'S-Video' 'Multi-Media'	'AV' 'DVI' 'Network'
OnScreenDisplay	'On'	'Off'	
PAPEnable	'Off'	'PIP'	'PBP'
PictureMode	'Standard' 'Custom'	'Vivid'	'Cinema'
PIPPosition	'Upper Left' 'Lower Right'	'Upper Right'	'Lower Left'
Power	'On'	'Off'	
Volume	0 to 100 in steps of 1		

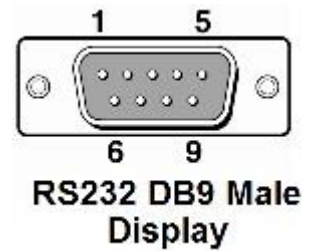
¹ Aspect ratio is Full for video and Full 2 for PC.² Aspect ratio is 4:3 for video and Real for PC.³ Aspect ratio is Wide Zoom for video and Full 1 for PC.

Cable and Adapter Requirements:

Captive Screw to Female DB9 RS-232 Serial Cable

Notes for the Device:**Serial communication:****Port Type:** RS-232**Baud Rate:** 9600**Data Bits:** 8**Parity:** None**Stop Bits:** One**Flow Control:** None**Pin Assignments Diagram**

Signal	Main Cable	Pin	Signal
TxD	→	2	RxD
RxD	←	3	TxD
GND	→	5	GND

**General Notes:**

Network communication:

When configuring the Ethernet interface, be sure the device settings match that of the GS interface settings.

Port Type:	Ethernet
Default Port:	4660
Logon Credentials Supported:	No
Multi-Connection Capabilities:	Undetermined
Port Changeability:	Undetermined

Ethernet Module Configuration Description:

Please refer to user manual for settings and changes to the network communication

Notes for the Device:

Appendix A. Set Commands

Aspect Ratio 4:3	801s1001\x0D
Aspect Ratio Full	801s1000\x0D
Aspect Ratio Wide Zoom	801s1002\x0D
Aspect Ratio Zoom	801s1003\x0D
Audio Mute Off	801s6000\x0D
Audio Mute On	801s6001\x0D
Auto Image None	801s\x8F\x0D
Executive Mode Off Type Button	801sE001\x0D
Executive Mode Off Type Button and IR	801sC001\x0D
Executive Mode Off Type IR	801sB001\x0D
Executive Mode On Type Button	801sE000\x0D
Executive Mode On Type Button and IR	801sC000\x0D
Executive Mode On Type IR	801sB000\x0D
Input AV	801s"003\x0D
Input DisplayPort	801s"007\x0D
Input DVI	801s"006\x0D
Input HDMI	801s"001\x0D
Input Multi-Media	801s"009\x0D
Input Network	801s"010\x0D
Input S-Video	801s"005\x0D
Input USB Display	801s"011\x0D
Input VGA	801s"000\x0D
Input YPbPr	801s"004\x0D
On Screen Display Off	801s[000\x0D
On Screen Display On	801s[001\x0D
PAP Enable Off	801s\x8A000\x0D
PAP Enable PBP	801s\x8A002\x0D
PAP Enable PIP	801s\x8A001\x0D
Picture Mode Cinema	801s\x81002\x0D
Picture Mode Custom	801s\x81003\x0D
Picture Mode Standard	801s\x81000\x0D
Picture Mode Vivid	801s\x81001\x0D
PIP Position Lower Left	801s\x8E002\x0D
PIP Position Lower Right	801s\x8E003\x0D
PIP Position Upper Left	801s\x8E000\x0D
PIP Position Upper Right	801s\x8E001\x0D
Power Off	801s!000\x0D
Power On	801s!001\x0D
Volume 0	801s5000\x0D
Volume 100	801s5100\x0D

Appendix B. Query Commands

Aspect Ratio	801gw000\x0D
Audio Mute	801gg000\x0D
Executive Mode Type Button	801gs000\x0D
Executive Mode Type Button and IR	801gi000\x0D
Executive Mode Type IR	801gh000\x0D
Input	801gj000\x0D
On Screen Display	801g]000\x0D
PAP Enable	801g\xBA000\x0D
Picture Mode	801g\xB1000\x0D
PIP Position	801g\xBF000\x0D
Power	801gl000\x0D
Volume	801gf000\x0D